

Fanambinantsoa Philibert ANDRINIRINIAIMALAZA

Dokotera-Injeniera · Mpikaroka Postdoctoral sy Mpampianatra

Elektrônika sy Informatika Industrialy, Rafitra Fanaraha-maso Manan-tsaina & Angovo Azo Havaozina Mifangaro

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MOMBA AHY (PROFILY AKADEMIKA)

Mpampianatra sady mpikaroka postdoctoral manana traikefa maherin'ny 10 taona amin'ny elektrônika sy informatika industrialy, rafitra fanaraha-maso manan-tsaina, elektrônika embarquée ary angovo azo havaozina mifangaro. Manam-pahaizana manokana amin'ny fanaraha-maso adaptative amin'ny alalan'ny fuzzy logic, ANFIS, PSO, ary teknika fanaraha-maso nomerika, ampiarina amin'ny rafitra PV/Rivotra/Bateria mifangaro, ny optimisation MPPT ary ny rafitra fitaovana fitaterana amin'ny fiara elektrika. Mpanoratra na mpiara-manoratra lahatsoratra siantifika maherin'ny 24 voadinika (peer-reviewed) sy fifandraisana IEEE (anisan'izany ny fandefasana roa ho an'ny IEEE EPEI 2026). Mpiara-miasa iraisam-pirenena mavitrika (Romania, Frantsa, Maorisy, Sina), miaraka amin'ny fahaiza-manao voamarina amin'ny fitarihana thèse (doctorat) any amin'ny andrim-panabeazana maromaro.

Vonona hiara-miasa ary mikaroka toerana ho Mpikaroka Postdoctoral, Mpampianatra, Maître de Conférences na Injenieran'ny R&D.

LAFY FIKAROHANA

- Fanaraha-maso adaptative sy manan-tsaina — Fuzzy logic, ANFIS, PSO, PID, MPPT, CNN
- Rafitra embarquée sy fotoana marina (temps réel) — PMSG, microcontrôleurs, fitaovana mifangaro
- Rafitra angovo azo havaozina — PV, Rivotra, Diesel, Bateria, turbine hydrolien
- Optimisation misy fepetra maro — vidin'ny angovo, entona CO₂, fahamendrehana (fampihenana LCOE)
- Élan elektrônika mahery (akademika, 2011–ankehitriny) — HVDC, MVDC, FACTS (SVC, STATCOM), famakafakana topolojia MMC, modelisation convertisseur mahery IGBT/IGCT, famolavolana fanaraha-maso ary famakafakana ny fahombiazana — novolavolaina sady nampianarina tao anaty thèse, taranja ary fandaharana injeniera.
- Simulation sy modelisation — MATLAB/Simulink, PSIM, LTSpice, LabView, XILINX ISE
- Governance ny microréseaux manan-tsaina ary fitantanana angovo mahaleo tena

FIANARANA

Vatsim-pikarohana Postdoctoral

Oniversite Teknika Gheorghe Asachi any Iași, Romania — Nov.–Des. 2024

Fikarohana momba ny paikady fanaraha-maso manan-tsaina ho an'ny convertisseur mahery ary ny fomba jumeau numérique ho an'ny rafitra elektrika.

Doctorat amin'ny Elektrônika Industrialy sy Informatika

Sekoly Doctorat EDT-EnRE, Oniversiten'i Antsiranana, Madagasikara — 2020

Thèse: Fandraisana anjara amin'ny fanatsarana ny fahombiazana angovon'ny fiara elektrika mifangaro. Famolavolana mpanaramaso Fuzzy–Sliding Mode ary PI gain miovaova ho an'ny fitaovana fitaterana maotera BLAC amin'ny fiara elektrika.

Fifindra-monina Doctorat

Oniversite Politehnica any Bucarest, Romania — Mey–Jol. 2016

Traikefam-pikarohana doctorat tao amin'ny Faculté du Génie Électrique, notohanana ny programme Eugen Ionescu an'ny AUF (Agence universitaire de la Francophonie). Fikarohana momba ny fanaraha-maso PI ordre fractionnaire sy fuzzy logic amin'ny fitaovana maotera PMSG/BLAC.

Diplaoman'Injeniera amin'ny Elektrônika Industrialy sy Informatika

École Supérieure Polytechnique d'Antsiranana (ESPA), Madagasikara — 2012

Tetikasa famaranana fianarana: Famolavolana sy fampiroboroboana fiara elektrika roa kodiarana mizaka.

TRAIKEFA AKADEMIKA

Mpampianatra sy Mpikaroka Postdoctoral | ISSTM – Oniversiten'i Mahajanga

2020 – Ankehitriny

- Taranja nampianarina (licence | master): Fizotry ny Famantarana (Signal Processing), Automatique, Faharanitan-tsaina Artifisialy, Fanaraha-maso Nomerika, Élan Elektrônika.
- Talen'ny licence amin'ny Génie Elektrônika sy Informatika Industrialy.
- Nitrika mihoatra ny 30 tetikasa famaranana fianarana sy mémoire Master 2.
- Lehiben'ny Sampandraharaha momba ny Fandraisana Mpianatra (nanomboka 2022).
- Teknisianina isan-tsokajy sy laboratoara.

Mpampianatra | FSTE – Oniversiten'i Mahajanga

2022 – Ankehitriny

- Taranja (licence | master): Elektrônika analogika sy nomerika, teknika fanaraha-maso mandroso (state feedback, adaptative, robuste, ary metaheuristika).

Mpampianatra Informatika | ESTI – Institut Supérieur des Technologies de l'Information

2020 – 2023

- Taranja (licence | master): PowerShell, fampiharana web mandroso (PWA), fandaharana Java miaraka amin'ny Spring Boot sy Spring MVC.

Mpampianatra sy Talen'ny Master | ESPA – Oniversiten'i Antsiranana

2015 – 2020

- Talen'ny Master amin'ny Telecommunications & Réseaux.
- Taranja (licence | master): Elektrônika analogika/numerika, teoria fanaraha-maso, rafitra microcontrôleur.
- Nitrika mihoatra ny 20 tetikasa famaranana fianarana sy mémoire Master 2.
- Teknisianina laboratoara.

Fampianarana Vonjimaika | Collège Saint-François-Xavier (CSFX), Fianarantsoa

2013 – 2014

- Fampianarana vonjimaika.

FITARIHANA FIKAROHANA

Fitarihana Doctorat Mavitrika — Doctorant 10 (Sekoly Doctorat 3)

DOCTORANT	LOHAHEVITRY THÈSE (FINTINA)	ANDRIM-PANABEAZANA / SEKOLY DOCTORAT
ANDRINIRINTSOA José M.	Fanombanana ny aloky ny PV amin'ny alalan'ny sky imager sy fianarana lalina multimodal	Oniv. La Réunion – ED-ST5
RANDRIAMIHELISOA Paolo R.	Fanaraha-maso ny famindrana angovo tsy misy tariby ho an'ny drone	Oniv. Mahajanga – EDGVM
SOAMIRAY Syllah B.Y.A.	Fitantanana drone mahaleo tena amin'ny alalan'ny interface ati-brain-computer	Oniv. Mahajanga – EDGVM

FREDERIC Moïse	Tambajotra sensor IoT ho an'ny fanaraha-maso toetr'andro (Boeny)	Oniv. Mahajanga – EDGVM
BOTOLAHY Angelo	Rafitra PV/Bateria mifangaro ho an'ny fitàna MADARAIL	Oniv. Antsiranana – EDT-EnRE
RANDRIA Amédé W.	Fampihenana ny fatiantoka herinaratra amin'ny tambajotra elektrifikasiona ambanivohitra	Oniv. Antsiranana – EDT-EnRE
DINY Oili	Fampidirana microréseaux manan-tsaina any an-tanàn-dehibe	Oniv. Antsiranana – EDT-EnRE
RABEARISON Bertho	Onduleur misy tariby mifandray amin'ny tambajotra tsy misy stockage — modelisation sy fanaraha-maso	Oniv. Antsiranana – EDT-EnRE
RAZAFINARIVO H.F.	Fitadiavana kilema manan-tsaina sy fanombanana famokarana ao amin'ny toby solar	Oniv. Antsiranana – EDT-EnRE
RANDJAKA Tsiloavy	Fitantanana angovo amin'ny rafitra PV/Bateria mifandray amin'ny tambajotra (ANFIS)	Oniv. Antsiranana – EDT-EnRE

Fitarihana Master Mavitrika — Mpianatra 3+ (sokajy: ISEA | GEI | GBM)

MPIANATRA MASTER	LOHAHEVITRA FIKAROHANA	MPIARA-MITARIKA
RANDRIAMANAFANA Alain	Famolavolana, simulation, ary fampiharana rafitra manan-tsaina mifototra amin'ny AI ho an'ny fanaraha-maso sy fitadiavana kilema ao amin'ny tambajotra elektrika	Prof. Nour M. Murad, Laboratoire PIMENT, Oniv. La Réunion ary Prof. MANASINA Ruffin, ISSTM Oniv. Mahajanga
AMBININTSOA Thierry Vincent	Fandrefesana antonony sy fitantanana angovo manan-tsaina ho an'ny nanoréseau CC an-trano mahaleo tena 100% : fampiharana amin'ny trano bioclimatique CC manontolo	—
RAKOTOARIMAMY Safidy Justine	Famolavolana sy fanatsarana surface manan-tsaina reconfigurable (RIS) mamokatra herinaratra ho azy manokana sady eco-conçue amin'ny plastika PET ho an'ny tambajotra Wi-Fi	—

FAHAIZANA TEKNIKA

- Simulation & Modelisation: MATLAB/Simulink, PSIM, LTSpice, LabView, XILINX ISE.
- Fitaovana rafi-pahereza: PSCAD, PSS® E, PowerWorld, GE-PSLF (thèse doctorat).
- Fandaharana (Programmation): Python, C, C++, Java, VHDL, C#
- Famakafakana Angona (Data Analysis): Python, R, IBM SPSS Statistics
- Famolavolana Hardware: Altium Designer, Eagle, FreeCAD, KiCAD
- CAO: AutoCAD, SketchUp, Blender
- Rafitra: PMSG, microcontrôleur embarqué, automatisation temps réel, convertisseur mahery
- Fiteny: Malagasy (teny ibiny), Frantsay (C1), Anglisy (B2 — Professional Working Proficiency)

ANDRAIKITRA & FANDRAISANA ANJARA

- Mpikambana amin'ny ekipa mpandrindra, Symposium Maintenance 2026, 11–12 Jona 2026, CNARP Antananarivo, Madagasikara (niarahan'ny ATIBM, IQLS, ISSTM, Mercy Ships nikarakara).
- Mpikambana amin'ny ekipa fandraisana EA-PGE (Photovoltaïque et Génie Électrique), Sekoly Doctorat Thématika momba ny Angovo Azo Havaozina sy Tontolo Iainana (EnRE), École Supérieure Polytechnique d'Antsiranana, Oniversiten'i Antsiranana (nanomboka Jona 2026).
- Tompon'andraikitra Fifandraisana (Communications Officer), ISSTM (nanomboka Janoary 2026).
- Mpampiofana, Fanoratana Siantifika ho an'ny Doctorant — Oniversiten'i Mahajanga (Des. 2025).

- Programa « Fampiofanana Mpampiofana » momba ny Fanoratana Siantifika — Tetikasa HayKa Erasmus+, Oniversiten'i Mahajanga (Aprily 2025).
- Lehiben'ny Sampandraharaha Partenariat sy Fiaraha-miasa, ISSTM (2022–2025).
- Mpikambana amin'ny Comité Scientifique, Oniversite Fahavaratra v.04, Oniversiten'i Mahajanga (2023).
- Mpandray anjara mavitrika: JRISTS (2021, 2022), Oniversite Fahavaratra (2021, 2023), Doctoriales (2019), Faha-40 taonan'ny FSTE (2024), AESMT'25, Faha-20 taonan'ny Oniversite Androna (2026), AESMT'26.
- Fiaraha-miasa fikarohana iraisam-pirenena: Romania, Frantsa (La Réunion), Maorisy, Sina (NUIST).

LAHATSORATRA AM-PANOMANANA & VOAFANTINA (24+ REHETRA)

Lahatsoratra am-panomanana (2026)

- Andriniriniainmalaza, F.P.; Murad, N.M.; Rhevihaja, S.N.; Andrinirintsoa, J.M.; Balan, G.; Manasina, R.; Lucache, D.D. (2026). *Marine and Hydraulic Energy Systems for Madagascar: Design, Optimization, and Prototyping*. Nalefa ho an'ny IEEE EPEI 2026, Iași, Romania.
- Andriniriniainmalaza, F.P.; Murad, N.M.; Randriamanafana, A.; Andrinirintsoa, J.M.; Balan, G.; Manasina, R.; Andrianirina, C.B.; Lucache, D.D. (2026). *Towards Low-Cost AI-Based Fault Monitoring and Diagnosis in Electrical Power Networks*. Nalefa ho an'ny IEEE EPEI 2026, Iași, Romania.
- Rakotomahefasoa, S.D.; Rhevihaja, S.N.; Andriniriniainmalaza, F.P.; Murad, N.M.; Andrinirintsoa, J.M.; Ravohitra, J.; Randriatefison, N.; Manasina, R. (2026). *Internal-pendulum wave energy converter: design, modeling, and experimental validation in Mahajanga*. Nalefa ho an'ny Journée Scientifique, ESPA, Oniversiten'i Antananarivo.
- Rhevihaja, S.N.; Randrianasolo, G.A.; Andriniriniainmalaza, F.P.; Murad, N.M.; Andrinirintsoa, J.M.; Ravohitra, J.; Randriatefison, N.; Manasina, R. (2026). *Numerical simulation optimization of a low-head Archimedes screw turbine: case study of a micro-hydropower plant in Madagascar*. Nalefa ho an'ny Journée Scientifique, ESPA, Oniversiten'i Antananarivo (fintina neken'ny mpandrindra).
- Soumah, D.; Murad, N.M.; Andriniriniainmalaza, P.; Rabearivony, A.F.; Georgiev, A.; Menard, S.G. (2026). *Predictive Design and Layer-by-Layer Characterization of 3D-Printed Multi-Layer Micro-Perforated Panels for Broadband Sound Absorption*. Fanitarana ny lahatsoratra AESMT'26; mbola eo am-panamarinana ho an'ny famoahana revy (RENE / EGY / ATE). Mpiara-manoratra.

Préimpression

1. Andriniriniainmalaza, F.P.; Randriamaitso, T.; Balan, G.; Rhevihaja, S.N.; Rakotomalala, L.; Lucache, D. (2025). *Hybrid Neuro-PSO MPPT control to improve the performance of vertical-axis hydrokinetic turbine columns*. SSRN Preprint. doi: [10.2139/ssrn.5856521](https://doi.org/10.2139/ssrn.5856521)
2. Andriniriniainmalaza, F.P.; Murad, N.M.; Bilal, H.; Khooaruth, A.; Randriatefison, N.; Ravelo, B. (2025). *Optimization of PV systems under shading faults using zone-based dynamic fuzzy logic and adaptive PSO*. SSRN Preprint. doi: [10.2139/ssrn.5229767](https://doi.org/10.2139/ssrn.5229767)

2026

1. Soumah, D.; Murad, N.M.; Andriniriniainmalaza, F.P.; Georgiev, A.G.; Mesnard, S.G. (2026). *Towards Broadband Acoustic Metamaterials for Tropical Noise Control: Design, Simulation and Experimental Characterization of FDM-Printed Micro-Perforated Panel Absorbers*. AESMT'26 (2 pages), Sofia, Bulgaria, 12–13 Mey 2026.

2025

1. Andriniriniainmalaza, F.P. et al. (2025). *Improvement and stabilization of the output voltages of a vertical-axis hydrokinetic turbine using intelligent control strategies*. IEEE ICECER 2025, Antananarivo, pp. 1–6. doi: [10.1109/ICECER65523.2025.11401321](https://doi.org/10.1109/ICECER65523.2025.11401321)
2. Andriniriniainmalaza, F.P. et al. (2025). *Hybrid fuzzy logic and shading-adaptive PSO for the dynamic mitigation of PV shading faults*. IEEE SIELMEN 2025, Iași, pp. 633–638. doi: [10.1109/SIELMEN67352.2025.11260746](https://doi.org/10.1109/SIELMEN67352.2025.11260746)
3. Rakotomalala, L.F.F.; Andriniriniainmalaza, F.P. et al. (2025). *Classification of daily load profiles for 20 kV distribution networks in Mahajanga, Madagascar*. IRJAES, Vol. 10, No. 3, pp. 86–92.
4. Balan, G.; Serea, E.; Andriniriniainmalaza, F.P.; Lucache, D.D.; Sălceanu, A.; Nenninger, P. (2025). *Digital twin approach for testing and control of automotive headlights*. ACEMP & OPTIM, Timișoara, pp. 1–7. doi: [10.1109/OPTIM-ACEMP62776.2025.11075229](https://doi.org/10.1109/OPTIM-ACEMP62776.2025.11075229)
5. Andriniriniainmalaza, F.P. et al. (2025). *Intelligent MPPT strategies for vertical-axis hydrokinetic turbines: comparative performance of Neuro-Fuzzy and Neuro-PSO controllers*. AESMT'25, Sofia, Bulgaria, 15–17 Mey 2025.

2024

1. Murad, N.M.; Andriniriniainmalaza, F.P. et al. (2024). *Study of a low-tech rainwater triboelectric generator as a renewable energy source*. IEEE EPEI 2024, Iași, pp. 626–631. doi: [10.1109/EPEI63510.2024.10758060](https://doi.org/10.1109/EPEI63510.2024.10758060)

2. Andriniriniimalaza, F.P.; Murad, N.M. (2024). *Neuro-Fuzzy control of a PMSG for a vertical-axis hydrokinetic turbine — output voltage improvement*. IEEE SESAI 2024, Mauritius, pp. 1–6. doi: [10.1109/SESAI61023.2024.10599409](https://doi.org/10.1109/SESAI61023.2024.10599409)
3. Andriniriniimalaza, F.P. et al. (2024). *Sizing methodology for a standalone PV/Wind/Diesel/Battery system minimizing LCOE and CO₂ emissions*. IJARIE, Vol. 10, No. 4, pp. 3400–3409.
4. Randriamaitso, T. et al. (2024). *Modeling and prediction of rainfall using the Neuro-Fuzzy method in the Marotandrano Special Reserve, Madagascar*. IJARIE, Vol. 10, No. 4, pp. 2775–2785.
5. Randriamaitso, T. et al. (2024). *Comparative study of the efficiency of optimal TSR and Neuro-Fuzzy MPPT regulation regarding torque impact on a PMSG-based VAHT*. IJARIE, Vol. 10, No. 3, pp. 6146–6158.

2022–2023

1. Randriamaitso, T. et al. (2023). *Study of the torque impact of a vertical-axis hydrokinetic turbine column on the electromechanical outputs of a PMSG*. AJSER, Vol. 6, No. 6.
2. Andrianantenaina, T.P.; Andriniriniimalaza, F.P. et al. (2023). *New direct nonlinear MPPT design for PV systems using backstepping and synergetic controllers*. IJMCR, 5(5), pp. 14–25.
3. Andriniriniimalaza, F.P. et al. (2022). *From classical control to hybrid fuzzy-logic strategies for brushless motor drives*. IJEI, 11(11), pp. 60–67.
4. Randjaka, T. et al. (2022). *ANFIS control approach for a bidirectional buck-boost converter for battery charging*. IJEMR, 12(3), pp. 30–39.
5. Andriniriniimalaza, F.P. et al. (2022). *Study of current drives applied to vehicle braking energy recovery*. WAJES, 9(3), pp. 1–6.
6. Andriniriniimalaza, F.P. et al. (2022). *Efficiency optimization approaches for hybrid electric vehicles*. IJERT, 11(5), pp. 326–331.

2017–2021 (safidy)

1. Randriamaitso, T. et al. (2021). *Study of torque impact on PMSG output voltages for a vertical-axis hydrokinetic turbine*. IJSET, 8(7), pp. 537–546.
2. Andriniriniimalaza, F.P. et al. (2019). *Fuzzy variable-gain PI control applied to AC brushless motor drives*. EAS, Vol. 4, No. 1, pp. 1–10.
3. Andriniriniimalaza, F.P. et al. (2018). *New hybrid Fuzzy-Sliding Mode control approach for AC brushless motors*. IJERT, 7(12), pp. 83–88.
4. Andriniriniimalaza, F.P. et al. (2017). *Fractional-order variable-gain PI control for a DFIG in a wind conversion chain*. IEEE ATEE 2017, Bucarest, pp. 740–745.
5. Andriniriniimalaza, F.P. et al. (2017). *Parameter optimization of a fuzzy-logic control for a brushless permanent-magnet motor*. IEEE ATEE 2017, Bucarest, pp. 211–216.

FANONDROANA (RÉFÉRENCES)

Prof. Dr. Ing. Dorin-Dumitru LUCACHE — Dean, Faculté de Génie Électrique, Oniversite Teknika Gheorghe Asachi any Iași, Romania — Fifandraisana azo alaina raha angatahina

Prof. Nour Mohamad MURAD — Laboratoire PIMENT, Oniversiten'i La Réunion, Frantsa — Fifandraisana azo alaina raha angatahina

Prof. Blaise RAVELO — Oniversite NUIST, Sina — Fifandraisana azo alaina raha angatahina